

## CHUGUYEVKA, Russian Federation

WMO: 319390

Lat: 44.150N

Lon: 133.867E

Elev: 849

StdP: 14.25

Time Zone: 10.00 (E10)

Period: 90-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -27.7      | -23.4 | -31.8                         | 0.9 | -27.2 | -27.2 | 1.2 | -22.9 | 12.2                  | 6.1  | 11.3 | 5.6  | 1.1                   | 120  | 0.503 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.3                   | 86.3            | 71.4 | 83.5 | 69.6 | 80.9 | 68.4 | 74.6                | 82.8 | 72.5 | 80.1 | 70.6 | 77.8 | 3.5                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 72.1                            | 122.4 | 79.7 | 70.1 | 114.1 | 77.0 | 68.1 | 106.6 | 74.8 | 38.9          | 82.3 | 36.9 | 79.8 | 35.2 | 78.0 | 81.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               |      | Max  |                            | Min   | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 12.0              | 10.0 | 8.5  | DB                         | -31.9 | 91.6 | 5.4                | 2.8                                                | -35.8     | 93.6 | -39.0      | 95.2 | -42.1      | 96.8 | -46.0      | 98.9 |
|                   |      |      | WB                         | -32.0 | 76.8 | 5.4                | 3.1                                                | -35.9     | 79.1 | -39.0      | 80.9 | -42.1      | 82.7 | -46.0      | 85.0 |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 37.5   | -3.6  | 8.0   | 24.0  | 40.2  | 53.9  | 63.2  | 70.3  | 68.5  | 56.8  | 41.5  | 22.2  | 3.1   |       |
|                                                                                  | DBStd                                                                            | 26.32  | 8.35  | 9.72  | 10.04 | 6.74  | 6.59  | 5.36  | 4.87  | 5.23  | 6.16  | 8.03  | 11.06 | 10.96 |       |
|                                                                                  | HDD50                                                                            | 6561   | 1660  | 1176  | 806   | 299   | 37    | 1     | 0     | 0     | 12    | 282   | 834   | 1454  |       |
|                                                                                  | HDD65                                                                            | 10399  | 2125  | 1596  | 1271  | 743   | 348   | 96    | 10    | 25    | 254   | 728   | 1284  | 1919  |       |
|                                                                                  | CDD50                                                                            | 2000   | 0     | 0     | 0     | 6     | 157   | 397   | 630   | 574   | 218   | 18    | 0     | 0     |       |
|                                                                                  | CDD65                                                                            | 363    | 0     | 0     | 0     | 0     | 3     | 42    | 175   | 134   | 9     | 0     | 0     | 0     |       |
|                                                                                  | CDH74                                                                            | 2982   | 0     | 0     | 0     | 3     | 124   | 483   | 1403  | 902   | 65    | 2     | 0     | 0     |       |
|                                                                                  | CDH80                                                                            | 764    | 0     | 0     | 0     | 0     | 23    | 98    | 418   | 216   | 9     | 0     | 0     | 0     |       |
| Wind                                                                             | WSAvg                                                                            | 3.0    | 2.1   | 2.8   | 3.7   | 4.3   | 3.8   | 3.0   | 2.9   | 2.4   | 2.5   | 3.1   | 3.3   | 2.3   |       |
| Precipitation                                                                    | PrecAvg                                                                          | 24.4   | 0.5   | 0.5   | 0.9   | 1.4   | 2.5   | 2.6   | 4.3   | 4.0   | 3.2   | 2.2   | 1.3   | 0.8   |       |
|                                                                                  | PrecMax                                                                          | 36.9   | 1.0   | 1.3   | 2.8   | 2.8   | 5.0   | 5.7   | 8.9   | 8.6   | 7.7   | 5.0   | 2.6   | 2.0   |       |
|                                                                                  | PrecMin                                                                          | 11.4   | 0.0   | 0.0   | 0.1   | 0.2   | 0.2   | 0.6   | 1.7   | 1.0   | 0.6   | 0.3   | 0.2   | 0.1   |       |
|                                                                                  | PrecStd                                                                          | 6.6    | 0.2   | 0.3   | 0.6   | 0.7   | 1.2   | 1.2   | 2.1   | 2.0   | 2.0   | 1.0   | 0.6   | 0.5   |       |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 25.2  | 40.6  | 56.0  | 72.2  | 84.5  | 86.4  | 90.9  | 89.8  | 81.1  | 70.7  | 54.0  | 37.0  |       |
|                                                                                  |                                                                                  | MCWB   | 22.9  | 34.2  | 43.0  | 52.4  | 60.4  | 66.6  | 75.3  | 74.1  | 69.2  | 57.1  | 45.3  | 34.1  |       |
|                                                                                  | 2%                                                                               | DB     | 19.9  | 34.2  | 49.9  | 66.4  | 78.2  | 82.8  | 87.6  | 85.0  | 74.8  | 64.5  | 47.4  | 32.5  |       |
|                                                                                  |                                                                                  | MCWB   | 17.8  | 29.7  | 39.6  | 48.9  | 59.3  | 66.1  | 73.3  | 72.1  | 63.9  | 53.2  | 42.0  | 30.5  |       |
|                                                                                  | 5%                                                                               | DB     | 15.1  | 29.5  | 44.3  | 59.8  | 73.7  | 79.8  | 84.5  | 81.9  | 71.5  | 60.0  | 42.4  | 25.2  |       |
|                                                                                  |                                                                                  | MCWB   | 13.4  | 26.3  | 36.0  | 46.1  | 56.0  | 64.9  | 71.2  | 70.7  | 62.0  | 50.7  | 37.8  | 23.5  |       |
|                                                                                  | 10%                                                                              | DB     | 11.4  | 25.4  | 40.1  | 55.3  | 69.7  | 76.2  | 81.6  | 79.0  | 68.5  | 55.8  | 37.8  | 20.9  |       |
|                                                                                  |                                                                                  | MCWB   | 9.9   | 22.4  | 33.1  | 43.8  | 54.8  | 64.0  | 70.2  | 69.5  | 60.5  | 48.4  | 33.8  | 19.2  |       |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 23.6  | 34.4  | 44.1  | 53.5  | 63.3  | 70.5  | 77.8  | 77.0  | 71.2  | 58.6  | 46.9  | 35.3  |
|                                                                                  |                                                                                  |        | MCDB  | 25.3  | 39.0  | 52.8  | 70.2  | 77.8  | 80.8  | 87.7  | 87.2  | 79.4  | 69.2  | 51.7  | 37.1  |
|                                                                                  |                                                                                  | 2%     | WB    | 17.9  | 30.4  | 39.7  | 49.7  | 60.5  | 67.7  | 75.4  | 74.3  | 66.3  | 54.7  | 42.4  | 29.7  |
|                                                                                  |                                                                                  |        | MCDB  | 19.6  | 33.6  | 48.4  | 63.2  | 74.5  | 79.3  | 84.4  | 81.7  | 71.5  | 61.6  | 46.9  | 32.2  |
| 5%                                                                               |                                                                                  | WB     | 13.5  | 26.4  | 36.3  | 46.9  | 58.1  | 66.1  | 73.4  | 72.4  | 63.9  | 51.9  | 38.4  | 23.9  |       |
|                                                                                  |                                                                                  | MCDB   | 14.8  | 29.3  | 43.6  | 58.6  | 69.5  | 77.0  | 81.5  | 79.4  | 69.1  | 58.4  | 42.5  | 25.2  |       |
| 10%                                                                              |                                                                                  | WB     | 9.9   | 22.3  | 33.6  | 44.2  | 55.9  | 64.5  | 71.4  | 70.5  | 61.7  | 49.0  | 34.2  | 19.1  |       |
|                                                                                  |                                                                                  | MCDB   | 11.3  | 25.4  | 39.1  | 54.0  | 66.9  | 75.0  | 79.2  | 76.9  | 67.1  | 54.9  | 37.0  | 20.8  |       |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 26.7  | 26.5  | 22.7  | 21.5  | 22.4  | 19.3  | 18.3  | 17.8  | 19.9  | 22.0  | 19.9  | 22.1  |
|                                                                                  |                                                                                  |        | MCDBR | 28.8  | 27.2  | 26.9  | 32.3  | 32.2  | 27.5  | 24.2  | 22.7  | 24.6  | 29.5  | 25.0  | 21.4  |
|                                                                                  | MCWBR                                                                            |        | 26.4  | 23.0  | 17.7  | 18.5  | 15.0  | 12.4  | 10.8  | 11.1  | 13.6  | 18.0  | 18.5  | 19.0  |       |
|                                                                                  | 5% WB                                                                            | MCDBR  | 28.3  | 26.7  | 24.9  | 30.9  | 27.3  | 24.5  | 21.5  | 19.8  | 19.0  | 26.4  | 23.0  | 20.4  |       |
|                                                                                  |                                                                                  | MCWBR  | 26.0  | 22.6  | 16.6  | 18.5  | 14.0  | 12.0  | 10.6  | 10.5  | 11.2  | 17.0  | 17.7  | 18.5  |       |
|                                                                                  | Clear-Sky<br>Solar<br>Irradiance                                                 | taub   |       | 0.317 | 0.381 | 0.463 | 0.561 | 0.567 | 0.540 | 0.501 | 0.465 | 0.422 | 0.433 | 0.382 | 0.338 |
| taud                                                                             |                                                                                  | 2.056  | 1.877 | 1.772 | 1.692 | 1.748 | 1.902 | 2.078 | 2.181 | 2.214 | 2.033 | 2.056 | 2.019 |       |       |
| Ebn at Noon                                                                      |                                                                                  | 243    | 243   | 238   | 226   | 231   | 238   | 246   | 251   | 252   | 225   | 215   | 218   |       |       |
| Edh at Noon                                                                      |                                                                                  | 33     | 48    | 61    | 72    | 70    | 61    | 50    | 44    | 39    | 41    | 34    | 31    |       |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           |        | 679   | 992   | 1302  | 1448  | 1601  | 1711  | 1544  | 1447  | 1294  | 938   | 642   | 542   |       |
|                                                                                  | RadStd                                                                           |        | 43    | 53    | 92    | 138   | 136   | 183   | 162   | 128   | 109   | 89    | 61    | 34    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (13 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page